

Meow Meow Call: Prototype Design for Building Interactive Connection between Human and Deaf Cat

Hongyi Zhang

Xi'an Academy of Fine Arts, China
90117@xafa.edu.cn

Sifu Zhu

Xi'an Academy of Fine Arts, China
972365412@qq.com

Yike Liu

Xi'an Academy of Fine Arts, China
kali.s@163.com

Jinjiang Ni

Xi'an Academy of Fine Arts, China
1206544875@qq.com

ABSTRACT

Animal-Computer Interaction (ACI) is a discipline that encourages study on animal interactions, and there are few studies that discuss disabled animals. This study will focus on cats with hearing loss and explore ways to build the emotional connection between humans and cats through interactions. The prototype design consisted of three parts, all of which were used in a two-week initial feasibility test. In the test with volunteers, two deaf cats and their keepers have participated in the interactive progress, and the owners' feedback showed some valuable results, which may refer to further studies around this topic.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design theory, concepts and paradigms;

KEYWORDS

Animal-Computer Interaction, deaf cat, interaction design

ACM Reference Format:

Hongyi Zhang, Yike Liu, Sifu Zhu, and Jinjiang Ni. 2021. Meow Meow Call: Prototype Design for Building Interactive Connection between Human and Deaf Cat. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451681>

1 INTRODUCTION

White cats with blue eyes are likely to be born with deafness is widely recognized in the public mind, and other factors (such as ototoxicity) can also cause hearing loss in cats [8]. The deafness generally will not cause any troubles to the pet cats because cats are frequently indoor animals only [8]. However, the communication and interaction between humans and cats may face some issues, which are due to the close of the auditory pathway in cats. Human

conveys meaning through talk, and feedback is important for human to build an emotional connection with animals. The study is inspired by a manifesto from ACI (Animal-Computer Interaction) [4], which suggested using technologies to foster the relationship between humans and animals by enabling communication and promoting understanding. Therefore, this study wishes to explore a way that may rebuild the emotional miss-link for both cats and humans. Based on the topic raised by ACI [4], the prototype design aims to develop deaf cats' life quality by building emotional connections with the keepers to fulfill their psychological needs. The prototype includes three main parts, the collar for the cat to wear, the mobile device for the human to operate, and the feeding machine that may create more opportunities for interaction. The interactive functions and procedures will be introduced in the following parts, and the initial feasibility test results will also be discussed.

2 RELATED WORK

Several studies work on ACI explored the principle of interaction [5, 9], the research methods in ACI [2], and interactive technology for bonding humans and animals [10]. The paper of Shaun Lawson et al. [22] argues that the use of technology to enhance interaction between humans and animals may damage animals' welfare and even the relationship between people and animals.

Besides, researchers have observed cat interactions at play [7, 13, 16, 19–21]. In 2009, Yonezawa et al. [16] studied attaching sensing units to pet cats to support human-pet interactions. Then J. Kim et al. [13] also proposed a concept of animal-robot interaction (ARI) which the human behalf robot will be attached to a cat to provide a service. The experiments showed the possibilities of implementation of ARI based on human-robot interaction studies. Besides, Noz and An [7] developed *Cat Cat Revolution*, a digital game of cat and mouse, and the pilot study explored the interaction between cat and owner during the game experiences. *Purrfect Crime* [21] also explored ACI's application in game design, which refers to the cat's instinct. That game allowed people and cats to participate simultaneously and provided a new way of inter-species gameplay. Pons et al. [19] discussed how *Felino* enhance the interaction between human and cat in the gameplay and adjusts the game by learning the cat's behavior and preferences. Pons [20] also proposed several suggestions to improve the inter-species gameplay based on the observational study of the cat's response to different technological stimulation.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451681>

In addition, Professor Strain has systematically studied Deafness in Dogs and Cats [8] and explained the auditory system, the form and mechanism of deafness, and how to live with a deaf dog or cat. Kral and Lomber [1] also specifically discussed the questions of the deaf white cat. Besides, Bradshaw's [15] research on how domestic cats make sense to their surroundings and his analysis of The Behavior of the Domestic Cat [14] has also enlightened interaction design ideas. The design idea of helping the owners build emotional communication with deaf cats is inspired by the previous studies on ACI, cat interactive games, and research of cat's nature.

3 CONCEPT

In addition to the previous studies, an introductory observational study with seven deaf cats and an open discussion with their owners (the owners mentioned in the following part in numbers 1 to 7). From the observational study, it shows that those deaf cats' daily life which is barely influenced by the loss of hearing, which match the study of Professor Strain [8] and Bradshaw [15], both sited that deafness will not influence domestic cats too much if they are kept indoor. However, the behaviors of deaf cats can be different from sane cats, and the keepers may feel emotionally disconnected from these deaf cats (agreed by owners 2,3,5,6, and 7). We sum up the main issues based on owners' statements and propose the ideas that inspired our design decisions in the following part:

The cats would be frightened by the touch, especially when the unexpected touch is out of their visual range stated by all the owners. Because the cats cannot be aware of the sounds, and any sudden physical contact may cause the stress response. It may suggest avoiding physical clues that may appear suddenly, such as vibration attached to the cat.

Cats who are born with deafness cannot understand humans communication through speech. Therefore they are unable to give any appropriate feedback. For instance, owner 2 has a kitten who was born deaf, and she finds "it is difficult to communicate with her," and the kitten's meow is loud "like it is angry." This misunderstanding of the cat's behavior may lead to a loss of emotional connection. The keepers (owners 3,5, and 6) also feel that the cat ignores their intention to communicate. In this case, the human becomes the starting point to solve the problem. The first step is to motivate the keepers to believe their cat can be emotionally reached and connected. In owner 1's case, she knows her cat is deaf and still keeps building a relationship with her cat. That cat has overcome the problem of silent communication and became closer with its keeper. However, not all the owners know their cats are deaf before they have them and be psychologically prepared for the upcoming issues.

The lack of communication in the long term may cause anxiety in both cats and humans. For example, owner 1 talked about her cat is following her all the round to ensure its owner was in sight, and she can tell that her cat will feel anxiety if she is away. Owner 4 has a cat who had been seriously ill. He complained that he is always worried about his cat because the cat cannot hear the sound of his calls after illness, and he is not used to the fact that his cat cannot respond to him as before.

All these problems may result in cats be abandoned, and it is difficult for deaf pet cats to survive in the wild. The study is concerned about the deaf cats' welfare and wishes to explore a new way of communication and interaction between humans and deaf cats through the idea of product design. This prototype design's main concept is to enhance the interactive connection between deaf cats and their keepers. Although, the cat is unable to hear the sound, it also means that sound will no longer affect the cat. In the other hand, there has to be another way to catch the cat's attention instead of vocal call to start a conversation between owner and cat. The design decision making will be based on both cat and human points of view and related scholars' findings. From the deaf cat's perspective, they can barely pick up the sounds from the environment and cannot understand any vocal signal or give any feedback. However, the other sense sensory organs of cats are complete, and the design will offer the other forms of signals rather than a voice for the deaf cats. From the keeper's point of view, human needs emotional connection and support from pets, but it can be a problem for the cats who are disabled to receive vocal calls. Therefore, the prototype design will create opportunities for interaction, and create reasonable virtual feedback to give keepers some psychological comfort. In addition, the design will focus on interactive needs based on daily life.

4 PROTOTYPE

Based on the concept, the prototype design will contain three parts, a mobile device for the human to send and receive signals, a collar with a small sphere device for cats to wear, and a food feeding machine. The ideas of product design are trying to match the natural sense of cat because Clutton-Brock [6] discussed that the cat has not evolved to obey human order. Therefore, the owner will take the lead in operating the interactive system, and the cat would not be asked to do anything against their nature. The diagram of the interaction between owners and cats is shown in Figure 1, which includes two goals as showing.

The collar with the sphere device (Figure 2) is designed as a medium that sends virtual feedback based on the keeper's calls, and the cat may also see a flashing light like a vision of "calling". The sphere device's diameter is 30mm, and it contains ATOM ECHO (that includes the sound device, RGB led lamp beads, Wi-Fi) and button battery. The model's weight is 22.64 grams, which is under the reference value proposed by Takasaki et al. [23]. In the study of wearable equipment for cat, Takasaki et al. [23] suggested that the weight of the cat's wearable equipment should be under 4~5% weight. In our test, the two cats weigh around 2.2kg to 3.5kg, which means the minimal weight can be up to 88g. As Bradshaw [15] stated, the cat was gifted in vision to detect tiny movements, and the deaf cat may be sensitive to other sensory stimuli modalities such as visual or floor vibration [8]. In this case, the light that will flash for 15 seconds each time may become a cue for cats to understand the human's calling. However, the light may not work if the cat cannot understand it as a signal of calling. In the interface design of keepers' mobile device (Figure 2), there is a button named "Find My Cat". The keeper can send out the request to the sphere device while they push the button, and the device will make a sound (like "Meow Meow") to acknowledge the keeper about their cat's

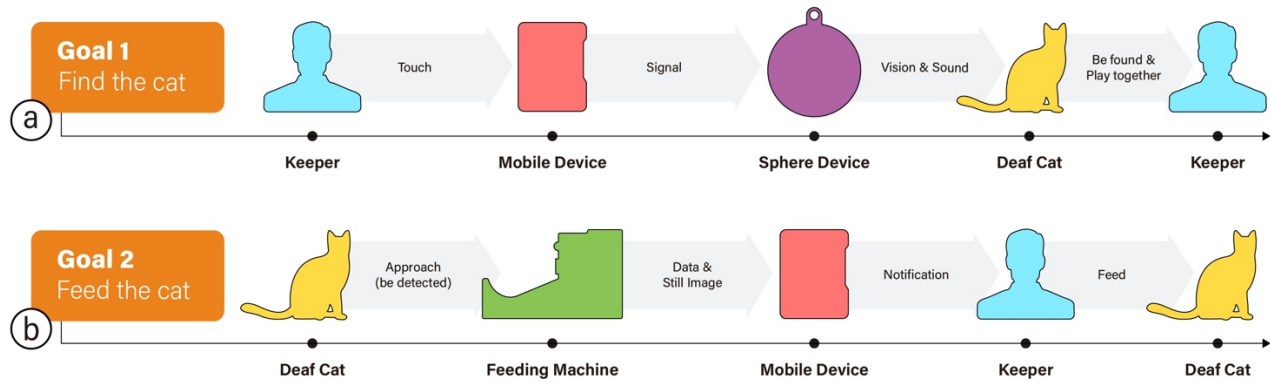


Figure 1: The diagram of the interaction methods based on different goals. A. Goal 1: Find the Cat, B. Goal 2: Feed the cat.

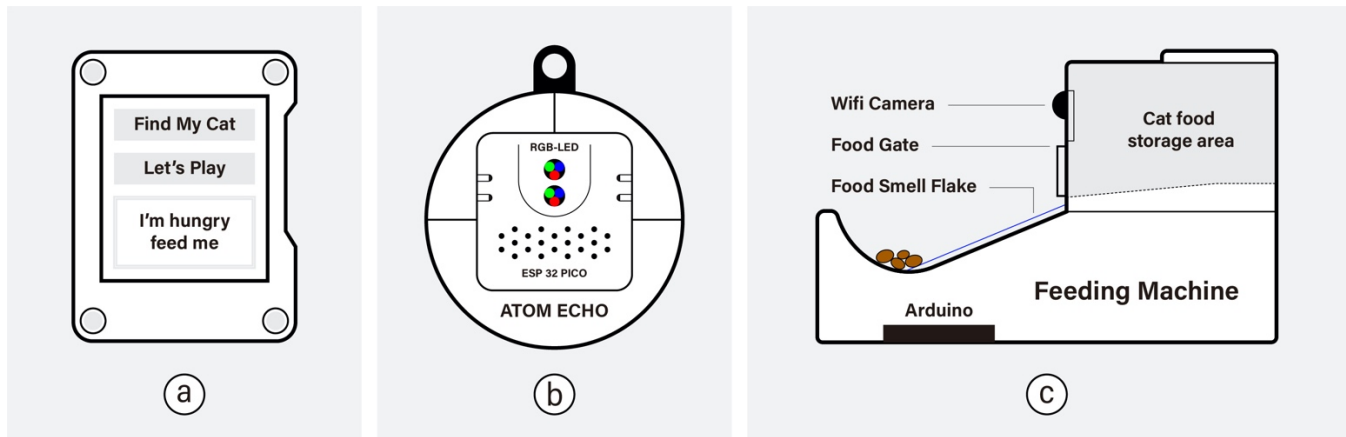


Figure 2: Interactive Devices, A. Mobile Device, B. Sphere Device, C. Feeding Machine.

location. In this case, the keeper will receive the virtual feedback that the cat will not be aware of, and this one-way interaction will not bother the deaf cat but satisfy the human need for emotional connection. On the other hand, if the keeper wishes to notice the deaf cat that they will play with them, there is a button named "Let's Play". Through the pathway similar to the previous one, the sphere device will make sounds and the flashing lights to grab the attention of deaf cats. Afterward, the deaf cat may learn this visual stimulation from the repeated interaction behaviors. Besides, the design decision of using blinking light instead of vibration is to avoid sudden physical contact. The light beads are set inside the sphere case, which is attached to the collar underneath the cat's chin. While the light is turned on, the cat will not look directly into the light, but it can see a flicker of refracted light on the ground instead. According to Peng's et al. [17] study of family LED lighting irradiation on eyes in mice experiment, they stated that LEDs are widely used in digital monitors and replaced light bulbs. They also proved that long blue light, which comes from a low-powered white family LED, causes retinal degeneration in mice after four weeks of the experiment. Besides, the previous experiments on cats using

the LED screen to play games (such as *Cat Cat Revolution* [7] and *Felino* [18]) show no specific explanation on whether LED screen may cause damage to the cat's eyes when the cat directly look at the tablet screen. However, the intensity of light and flashing duration were all considered not to disturb the cat, and the research in this area is lacking. Two colors of light were chosen, red and blue. Refer to Bradshaw's et al. [14] study, cat eyes can only be sensitive to two types of lights, which are green and blue. The lack of red-sensitive cones to the cat means it cannot see the color red, and the red objects will appear much darker than green and blue objects. Based on this statement of Bradshaw [14], we offered a "bright" light (blue) and a "dim" light (red) to see the cat's reaction.

The feeding machine's size (Figure 2) is 20.5cm in height, 21cm in width, and 32cm in-depth, and the machine has a built-in surveillance Camera, Arduino UNO, and other related units. It was designed based on the cat's foraging nature, which is used smell sense to identify food [15]. A food smell flake has been placed in the feeding bowl section, and it will attract the hungry cat to come to the machine. The cats' searching and smell motions will be captured and transferred to the mobile device as a signal that reminds

people to feed their cats. In addition, domestic cats' studies point out that sociality as part of domestication relied on concentrations of food supplied by man [14]. It means the cat will develop relationships with people who feed them every day, and this is why the design considered starting with a feeding machine. Ito et al. [12] suggested that humans may develop a positive relationship with cats if they can interact with them by orienting their faces and call in faces cats' names. In addition, Chijiwa et al. [3] explored that cats can be attracted by human eating behavior and attend to the food container. It means the premise of food can offer a positive opportunity to interact with cats. Therefore, the interaction begins with capturing or arousing the cat's desire to eat. As the surveillance camera detected the cat has approached the feeding area, it will start recording and sending a notification to the owner's mobile device simultaneously. The food storage gate will be opened and release about 5g cat snacks, while the Arduino UNO recognized the signal of the cat's attendance. The cat will pay attention to the snack for a while, and the keeper can own some time to prepare a real meal for the cat. As this procedure repeats in their daily life, the cat may realize their approach is the trigger to gaining meals and their keeper's attention. Based on the previous interaction mechanism, the keeper can be more involved in the deaf cat's feeding process, and this will help to strengthen their relationship bond. To test out the function, structure, and interactive mechanism of the product design ideas, the study has set up the following experiment to explore a potentially new perspective of human and deaf cats' interaction.

5 INITIAL FEASIBILITY TEST

An initial feasibility test was held to verify whether the product model can meet expectations. There are two cats (Cat A and Cat B) and their owners were involved. Based on the previous ethic study by Heli Väättäjä et al. [11], all the voluntary owners can withdraw from the test at any time if they feel or think that the test will affect the cat's physical and mental health. The owners would be also fully acknowledged the content and process of the test before the start. During this initial feasibility test process, the cats lived in their original environment, although the test might be an informal experiment. The reason for holding the test in their original living environment is to ensure their comfort.

Cat A is a 6-month-old, female, in the weight of 2.2kg. It lives with two owners, one female and one male, but it was mainly the hostess who took care of its life. Cat A lives in a house with two bedrooms and one living room. It likes to play in the living room, and hides in one of the bedrooms for rest. Cat B is 4-years-old, male, in the weight of 3.5kg. It lives with a male owner in a house with an entrance, two bedrooms, one living room, and a balcony. It normally stays at the entrance or stays at the balcony for relaxation. Andrej and Stephen [1] cited that the general forms of hearing loss include conductive, sensory, or neuronal. In this condition, Cat A was born with deafness, and Cat B lost hearing from ototoxic drugs.

In the test period, the cats requested to wear the collar with the attached sphere device, and the food machine had also been introduced to their daily life. The food machine will be placed in their original feeding locations, where is in the corner of the living-room of Cat A's house, and where is in the entrance of Cat

B's house. At the beginning, the owners would spend some time understanding how to manipulate the equipment and achieve basic interaction goals. The goals include feeding the cats at least 2 times per day, running "Find My Cat" at least 3 times per day, and play with the cat at least 2 times per day with "Let's Play". The keepers were asked to observe the cats' reactions, record the activities, as well as their thoughts or feelings. However, the participants are allowed to stop the test or skip some request if the cat does not cooperate or feel unnecessary. If they skipped some request, they need to report what has been skipped and why they have to do so.

At the end of the test, the two owners were invited to evaluate the use of these two product models and openly discuss and share their experiences, records and feelings. As result (Figure 3), there are different performances among these two cats.

In Cat A's case, the keeper found their emotional bond has become stronger, because the feeding machine offered more chances for her to participate in the cat's mealtime. It was hard to build communication through feeding activity because the cat cannot hear the calls and would not show up when the food had placed. On the other hand, Cat A showed intense attention to the flashing lights, and enjoy playing with the lights at the beginning. The owner stated that she almost followed the request, except at the first 2 days she can only use "play with me" once at evening, because she was too busy to do it again during her free-time. She also suggested the feeding machine's height, which was a little bit too high for her kitten. "It was okay for the kitten to reach the food through side-way, but the front edge seems to be a barrier for her."

In Cat B's case, the cat will call the keeper for food as usual, and the feeding machine only provides an extra notification and record to the keeper. "It's interesting to have a picture of cat eating from that angle, I'd like to make an album of it. But I don't think I'll use this machine to replace my old automatic feeder, or maybe you can combine them." However, the calling program on the mobile device is more helpful for the owner. Although Cat B is less interested in the flashing lights, the owner think is playful to see his cat seeing this visual communication signal. "Cat B has become quiet after the illness. Sometimes I wonder if he gets bored because he can't hear the sound anymore. I think perhaps he knew the trick of the light, because when the light shows up he may look around like finding something." The owner claimed the virtual sound feedback also makes him feel reassuring about the cat. "Sometimes I do check where Cat B is according to the location of the sound, and see what he's doing."

Based on the purpose of testing the product function and mechanism, we conducted this two-week informal test. We did not ask the participants to strictly complete the test, which may affect the participant cats' welfare. Based on the keepers' feedback, they both found a new way to interact with their cats, and it also gives cats some clues about human behaviors. The positive feedback encouraged us to develop new ways of communicating between humans and deaf cats and further research interspecies communication and ACI.

6 CONCLUSION

Referring to the research on ACI, this prototype design explores the possibility of improving the interaction between deaf cat and



Figure 3: Cats Interaction Records.

keepers. The concept of the prototype is to rebuild the miss-linked connection between deaf cats and their keepers. There are three devices have been designed. The collar with a sphere device can give the keeper sound feedback and shows a flashing light as a “vision call” to the deaf cat. The feeding machine can attract cat to approach, and the build-in surveillance camera will detect cat attendance and send a notification to the keeper. The mobile device used for the keeper, who can send signals of finding the cat and receive notice from the feeding machine. All the products were designed based on studies of cat’s nature. In addition, the initial feasibility test with two cats and their owners showed some expected results. For example, the owner claimed their relationships with the cats were enhanced through interactive feeding behavior, and the call’s virtual feedback will make the keeper feel more comfortable. However, the experiment’s limitation may not reflect more comprehensive problems, and the rough appearance of the prototypes may affect the user’s experience. Besides, the participants were allowed to skip some of the requests if the cats or owners had no willingness to do it. Therefore, the scale of formal experiments should expand, and the number of samples should increase to find more possibilities. In the future, the prototype may be iterated based on further studies of user experiences, and different types of interactive devices maybe designed to create new ways of interaction.

REFERENCES

- [1] Andrej Kral, Stephen G. Lomber. 2015. Deaf White Cats. *Curr. Biol.* 25 (9), R351–R353. <https://doi.org/10/ghrq5w>.
- [2] Anna Zamansky, Amanda Roshier, Clara Mancini, Emily C. Collins, Carol Hall, Katie Grillaert, Ann Morrison, Steve North, and Hanna Wirman. 2017. A Report on the First International Workshop on Research Methods in Animal-Computer Interaction. In *CHI’17 Extended Abstracts on Human Factors in Computing Systems*, pp. 806–815. ACM. <https://doi.org/10.1145/3027063.3052759>.
- [3] Chijiwa, H., Takagi, S., Arahori, M., Hori, Y., Saito, A., Kuroshima, H., & Fujita, K. 2020. Dogs and cats prioritize human action: choosing a now-empty instead of a still-baited container. *Animal Cognition*, 1–9. <https://doi.org/10.1007/s10071-020-01416-w>.
- [4] Clara Mancini. 2011. Animal-computer interaction: a manifesto. *Interactions* 18.4: 69–73. <https://doi.org/10.1145/1978822.1978836>.
- [5] Clara Mancini. 2013. Animal-computer interaction (ACI): changing perspective on HCI, participation and sustainability. In *CHI’13 Extended Abstracts on Human Factors in Computing Systems*, pp. 2227–2236. ACM. <http://doi.org/10.1145/2468356.2468744>.
- [6] Clutton-Brock J. 1988. *The British Museum book of cats: ancient and modern*. British Museum Press, London.
- [7] Frank Noz and Jinsoo An. 2011. Cat cat revolution: an interspecies gaming experience. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI ’11)*. Association for Computing Machinery, New York, NY, USA. pp. 2661–2664. Doi: 10.1145/1978942.1979331.
- [8] George M. Strain. 2011. *Deafness in Dogs and Cats*. CABI: Cambridge, MA.
- [9] Heli Väättäjä, Emilia K. Pesonen. 2013. Ethical issues and guidelines when conducting HCI studies with animals. In *CHI’13 Extended Abstracts on Human Factors in Computing Systems*, pp. 2159–2168. ACM. <http://doi.org/10.1145/2468356.2468736>.
- [10] Heli Väättäjä, Päivi Majaranta, Heini Törnqvist, Mari Ainasoja, Veikko Surakka, Oskar Juhlin, and Clara Mancini. 2017. Technology for Bonding in Human-Animal Interaction. In *Proceedings of the Fourth International Conference on Animal-Computer Interaction*. Association for Computing Machinery: New York, NY, USA, 2017. pp. 1–5. <https://doi.org/10.1145/3152130.3152153>.
- [11] Heli Väättäjä, Päivi Majaranta, Poika Isokoski, Yulia Gizatdinova, Miiamaaria V. Kujala, Sanni Somppi, Antti Vehkaoja, Outi Vainio, Oskar Juhlin, Mikko Ruohonen, and Veikko Surakka. 2018. Happy dogs and happy owners: using dog activity monitoring technology in everyday life. In *Proceedings of the Fifth International Conference on Animal-Computer Interaction (ACI ’18)*. Association for Computing Machinery, New York, NY, USA, Article 9, pp. 1–12. <https://doi.org/10.1145/3295598.3295607>.
- [12] Ito, Y., Watanabe, A., Takagi, S., Arahori, M., & Saito, A. 2016. Cats Beg For Food From The Human Who Looks At And Calls To Them: Ability To Understand Humans’ attentional States. *Psychologia*, 59(2-3), pp. 112–120. Doi: 10.2117/Pyssoc.2016.112.
- [13] J. Kim, S. Choi, D. Kim, J. Kim and M. Cho. 2009. Animal-Robot Interaction for pet caring. 2009 IEEE International Symposium on Computational Intelligence in Robotics and Automation - (CIRA), Daejeon. pp. 159–164. Doi: 10.1109/CIRA.2009.5423214.
- [14] John Bradshaw. 2012. *The Behaviour of the Domestic Cat*. CABI.
- [15] John Bradshaw. 2013. *Cat Sense: How the New Feline Science Can Make You a Better Friend to Your Pet*. Basic Books: New York.
- [16] Kyoko Yonezawa, Takashi Miyaki, and Jun Rekimoto. 2009. Cat@Log: Sensing Device Attachable to Pet Cats for Supporting Human-Pet Interaction. In *Proceedings of the International Conference on Advances in Computer Entertainment Technology; ACE ’09*. Association for Computing Machinery: New York, NY, USA. pp. 149–156. <https://doi.org/10/cg5cx3>.
- [17] Mei-Lin Peng, Cheng-Yu Tsai, Chung-Liang Chien, John Ching-Jen Hsiao, Shuan-Yu Huang, Ching-Ju Lee, Hsiang-Yin Lin, Yang-Cheng Wen & Kuang-Wen Tseng. 2012. The Influence of Low-powered Family LED Lighting on Eyes in Mice Experimental Model. *Life Science Journal*. 2012;9(1):477–482 [ISSN:1097-8135]. <http://www.lifesciencesite.com>.
- [18] Michelle Westerlaken and Stefano Gualeni. 2014. Grounded Zoomorphism: An Evaluation Methodology for ACI Design. *Proc. 2014 Work. Adv. Comput. Entertainment Conf. - ACE ’14 Work*. pp. 1–6. <https://doi.org/10.1145/2693787.2693796>.
- [19] Patricia Pons, Javier Jaen, and Alejandro Catala. 2015. Envisioning Future Playful Interactive Environments for Animals. In *More Playful User Interfaces: Interfaces that Invite Social and Physical Interaction*; Nijholt, A., Ed.; Gaming Media and

- Social Effects; Springer: Singapore. pp. 121–150. https://doi.org/10.1007/978-981-287-546-4_6.
- [20] Patricia Pons, Javier Jaen. 2016. Towards the Creation of Interspecies Digital Games: An Observational Study on Cats' Interest in Interactive Technologies. In CHI'16 Extended Abstracts on Human Factors in Computing Systems. Association for Computing Machinery: New York, NY, USA. pp. 1737-1743. <https://doi.org/10/ghrqf>.
- [21] Rui Trindade, Micaela Sousa, Cristina Hart, Nádia Vieira, Roberto Rodrigues, and João França. 2015. Purrfect Crime. Proc. 33rd Annu. ACM Conf. Ext. Abstr. Hum. Factors Comput. Syst. - CHI EA '15, pp. 93–96. <https://doi.org/10.1145/2702613.2728660>.
- [22] Shaun Lawson, Ben Kirman, Conor Linehan, Tom Feltwell, and Lisa Hopkins. 2015. Problematising Upstream Technology through Speculative Design: The Case of Quantified Cats and Dogs. In Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems; CHI '15; Association for Computing Machinery: New York, NY, USA, 2015. pp. 2663–2672. <https://doi.org/10/ghrqh>.
- [23] Takasaki T, Nagato R, Morishita E, *et al.* 2001. Wild animal tracking by PHS (Personal Handy phone System). Human Friendly Mechatronics. Elsevier Science. pp. 43-48. <https://doi.org/10.1016/B978-044450649-8/50009-7.</bib>>